Alessandro **Sanvito**

SOFTWARE AND AI ENGINEER

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Experience

Optiver *Amsterdam, Netherlands*

SOFTWARE ENGINEER *August 2023 - Current*

- Built baseline systematic strategies for new asset classes from scratch, including data sourcing, cleaning, and validation, for researchers to iterate on.
- Selected candidate signals and defined the validation criteria each baseline must meet before handoff.
- Cut backtest runtime and migrated ML model training from local to cloud with Python, Spark, Polars, DuckDB, PostgreSQL, and Delta Lake.

Mercedes-Benz (AI-SEE Project) *Stuttgart, Germany*

AI RESEARCH INTERN AND MASTER THESIS STUDENT *May 2022 - June 2023*

- Researched 3D human avatar modeling from monocular video, building generative neural models (NeRFs, Diffusion Models) to improve computer vision in adverse weather.
- Collaborated with a cross-functional R&D team, resulting in a first-author publication at IEEE IV 2024 and a co-authored publication at ICCV 2023.

Education

KTH Royal Institute of Technology *Stockholm, Sweden*

MSC. IN ICT INNOVATION, DATA SCIENCE *Sept. 2020 - May 2023*

- Final grade: A - Excellent
- Implemented from scratch a diverse range of neural network architectures in NumPy, including MLPs, Hopfield Networks, and Deep Belief Networks.

Polytechnic University of Milan *Milan, Italy*

MSC. IN COMPUTER SCIENCE AND ENGINEERING *Sept. 2020 - Jul. 2023*

- **1st place (academic)** at the international ACM RecSys Challenge 2021: co-developed the winning ML solution on the Twitter engagement dataset with the university team under prof. Paolo Cremonesi, using Dask, CatBoost, and XGBoost.
- Final grade: 110/110 cum laude

Polytechnic University of Milan *Milan, Italy*

BSC. IN ENGINEERING OF COMPUTING SYSTEMS *Sept. 2017 - Jul. 2020*

- Final grade: 109/110

Publications

2024	**HINT: Learning Complete Human Neural Representations from Limited Viewpoints**, **Sanvito Alessandro** (first author), et al.	*IEEE IV 2024*
2023	**ScatterNeRF: Seeing Through Fog with Physically-Based Inverse Neural Renderings**, Ramazzina Andrea, et al., incl. **Sanvito Alessandro**	*ICCV 2023*
2022	**United We Stand, Divided We Fall: Leveraging Ensembles of Recommenders to Compete with Budget Constrained Resources**, Maldini Pietro, **Sanvito Alessandro**, Surricchio Mattia	*ACM recsys'22*
2021	**Lightweight and Scalable Model for Tweet Engagements Predictions in a Resource-constrained Environment**, Carminati Luca, et al., incl. **Sanvito Alessandro**	*ACM recsys'21*

Skills

Programming	Python, SQL, Java, C++, C
Research & ML	PyTorch, generative models (NeRFs, Diffusion), computer vision, recommender systems, gradient boosting (CatBoost, XGBoost), scikit-learn
Data Engineering	Spark, Polars, DuckDB, PostgreSQL, Delta Lake, Pandas, NumPy, MLlib, cloud compute
Tools & Platforms	Git, Docker, Linux, CI/CD, pytest